

Genomics of Drug Sensitivity in Cancer (GDSC): Analysing Rapamycin (mTOR Inhibitor) Sensitivity in Blood and Solid Cancers

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Introduction

There are many types of cancer drugs that differ greatly in their mechanisms. This project focused on a specific drug called rapamycin, which blocks mTORC1, part of the general mTOR-directed growth pathway[1]. mTORC1 is signalling complex that is very important in cell proliferation and division, and since this is often the basis of tumours, it makes rapamycin interesting to analyse as a cancer drug. Blood cancers are often reported to respond strongly to growth-signalling inhibitors, but it is unclear whether this reflects something specific about mTOR inhibition or a general feature of blood cancer cells. This project therefore asked: *are blood cancers more sensitive to rapamycin than solid tumours, and is this difference larger than for drugs in general?*

Dataset and Method

The dataset used was the Genomics of Drug Sensitivity in Cancer 2 drug screen [2], in which hundreds of cancer cell lines were tested against hundreds of drugs. Sensitivity was measured by the IC50. This corresponds to the dose that halves cell growth [3]. The dataset stored this value on the natural-log scale (LN_IC50), where a lower value means a more sensitive cell line. For rapamycin, repeated measurements were averaged to one value per cell line and unclassified lines were removed. The remaining lines were grouped by TCGA cancer type into 111 blood cancer lines (Table 1) and 655 solid tumour lines. The difference in group means was tested with Welch's t-test which does not assume equal variances or group sizes. The same comparison was repeated for all 286 drugs to test whether rapamycin is special, or whether it displays similar trends to other drugs. The analysis is a single Python script (`rapamycin_analysis.py`) using pandas, numpy, scipy and matplotlib.

TCGA type	Cell lines (n)	Disease
DLBC	31	Diffuse large B-cell lymphoma
ALL	26	Acute lymphoblastic leukemia
LAML	25	Acute myeloid leukemia
MM	17	Multiple myeloma
LCML	10	Chronic myelogenous leukemia
CLL	2	Chronic lymphocytic leukemia
Total	111	

Table 1: Blood cancer cell lines with a rapamycin measurement, by TCGA cancer type.

Results and discussion

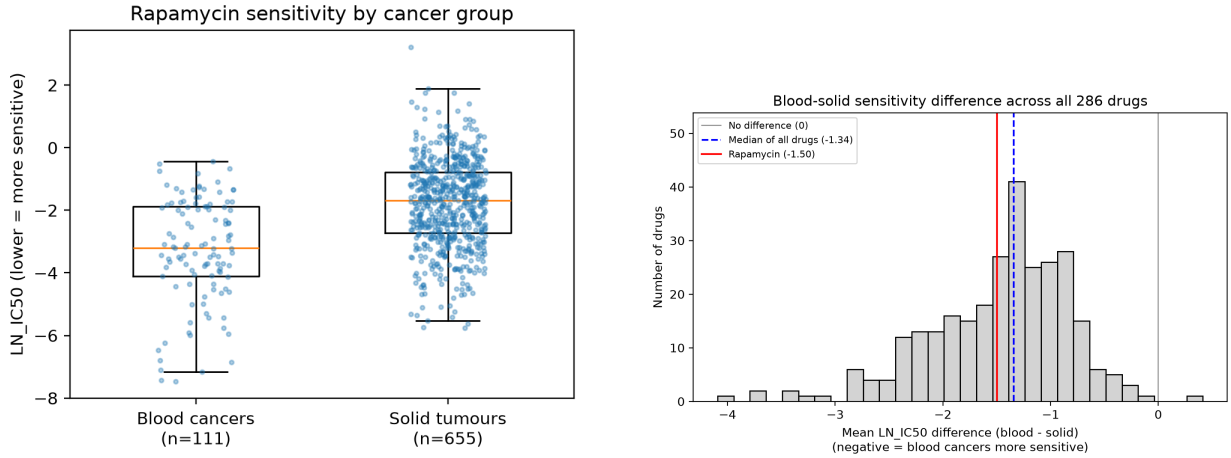
Rapamycin analysis: Blood cancer lines were significantly more sensitive to rapamycin than solid tumour lines (Figure 1a). The mean LN_IC50 was -3.28 ± 0.16 in blood lines and -1.78 ± 0.05 in solid lines (mean $\pm$ standard error), a difference of -1.50 ± 0.17, or about 4.5-fold lower IC50 in blood lines. Welch's t-test gave $t = 8.9$ and $p \approx 3 \times 10^{-15}$.

	Blood cancers	Solid tumours
Cell lines (n)	111	655
Mean LN_IC50 ($\pm$ Standard Error)	-3.28 ± 0.16	-1.78 ± 0.05

Table 2: Rapamycin sensitivity by cancer group. Difference in means: -1.50 ± 0.17; Welch's t-test $t = 8.9$, $p \approx 3 \times 10^{-15}$.

All drugs: However, the mean difference between blood and solid cancers followed the same trend for 285 of 286 drugs (Figure 1b). Rapamycin ranked 117th of 286 drugs. The median difference across drugs was -1.34, and the middle 50% of drugs spanned -1.88 to -1.03, so rapamycin's -1.50 lies well inside this range.

The blood lines responded more strongly to almost every compound tested, and there was no evidence that mTOR inhibition is more effective in blood cancers.



(a) (b)

Figure 1: (a) Rapamycin LN_IC50 in blood cancer (n=111) and solid tumour (n=655) cell lines. (b) Blood-minus-solid difference in mean LN_IC50 for all 286 drugs, with rapamycin (red) and the median of all drugs (dashed blue).

Limitations: Blood lines tended to respond more to drugs in general, so a group comparison cannot separate drug-specific effects from this baseline. Blood cancers are not one group either, as the 111 blood lines are six TCGA cancer types[4] (Table 1), with CLL represented by only two lines, so these subgroups should be analysed separately. Finally, cell lines are not patients, they do not have an immune system or a blood supply so the results do not translate clinically.

Conclusion

Blood cancer cell lines are about 4.5 times more sensitive to rapamycin than solid tumour lines. However, 285 of the 286 drugs displayed the same trend, and rapamycin's difference sat inside the middle half of all drugs. The answer to the question is that blood cancers are more sensitive to rapamycin, but not more than they are to drugs in general, so this result alone does not point to a rapamycin or mTOR specific effect. Future analysis could involve correcting for each cell line's general drug sensitivity, comparing drugs that act on the same pathway, and analysing blood sub-groups separately.

References

[1] J. Li, S. G. Kim, and J. Blenis. Rapamycin: One Drug, Many Effects. *Cell Metabolism*, 19(3):373–379, 2014. doi:10.1016/j.cmet.2014.01.001.

[2] Genomics of Drug Sensitivity in Cancer (GDSC) *GDSC2 Dataset*. Available at: https://www.kaggle.com/datasets/samiraalipour/genomics-of-drug-sensitivity-in-cancer-gdsc.

[3] S. Aykul and E. Martinez-Hackert, "Determination of half-maximal inhibitory concentration using biosensor-based protein interaction analysis," *Anal. Biochem.*, vol. 508, pp. 97–103, Sep. 2016. doi: 10.1016/j.ab.2016.06.025. PMID: 27365221; PMCID: PMC4955526.

[4] National Cancer Institute Genomic Data Commons (GDC). *TCGA Study Abbreviations*. Available at: `https://gdc.cancer.gov/resources-tcga-users/tcga-code-tables/tcga-study-abbreviations`. Accessed: October 1, 2026.